EXPERIMENT NO 6

Use Figma to simulate a handoff scenario in a mobile network and visualize the process.

Aim: -To simulate a handoff scenario in a mobile network and visualize the process using Figma.

Procedure:

1. Open Figma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation
5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia
9. Storyboard Animation
10. Review and edit the Prototype
11. Save and Share

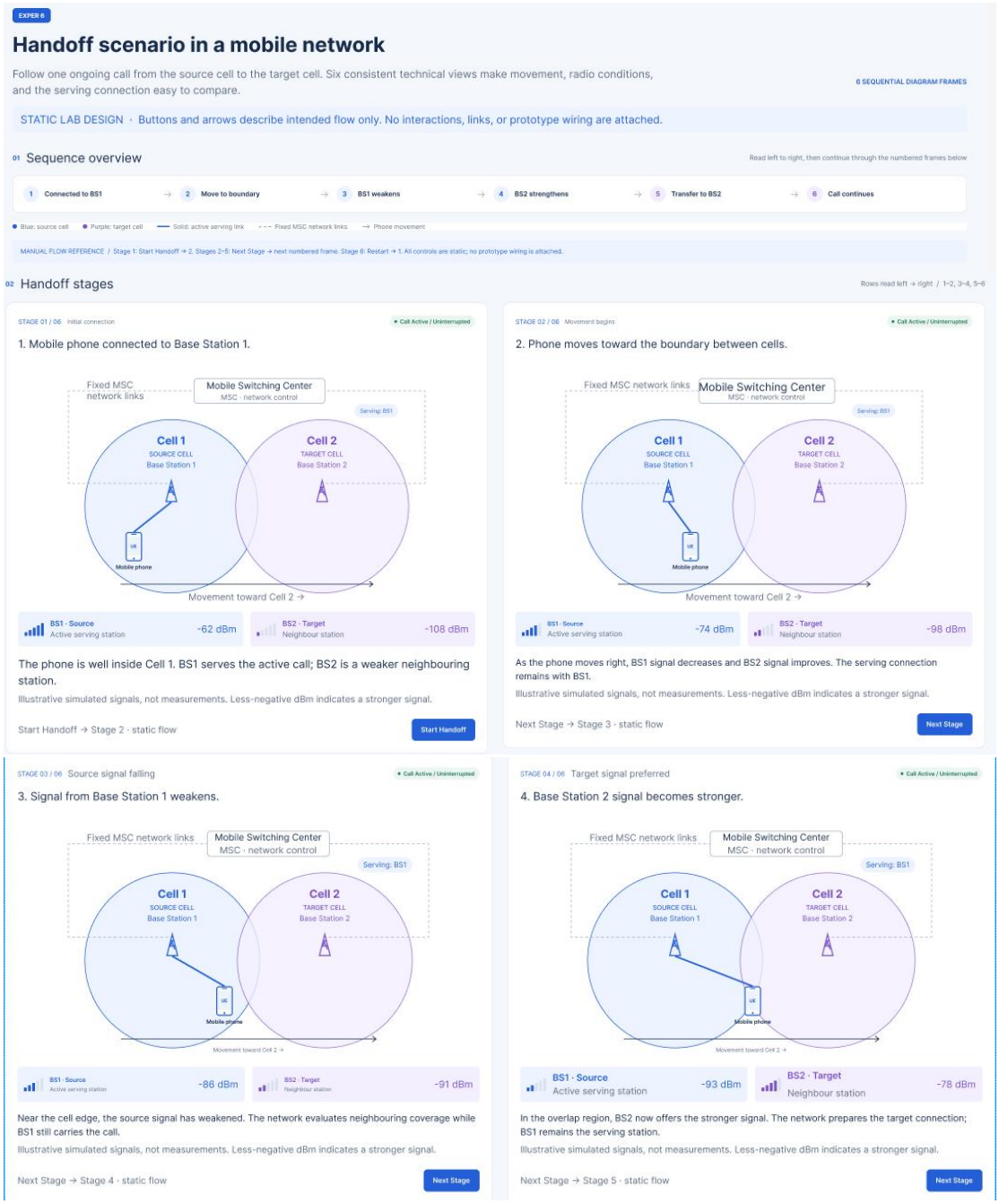
Components and Their Functions

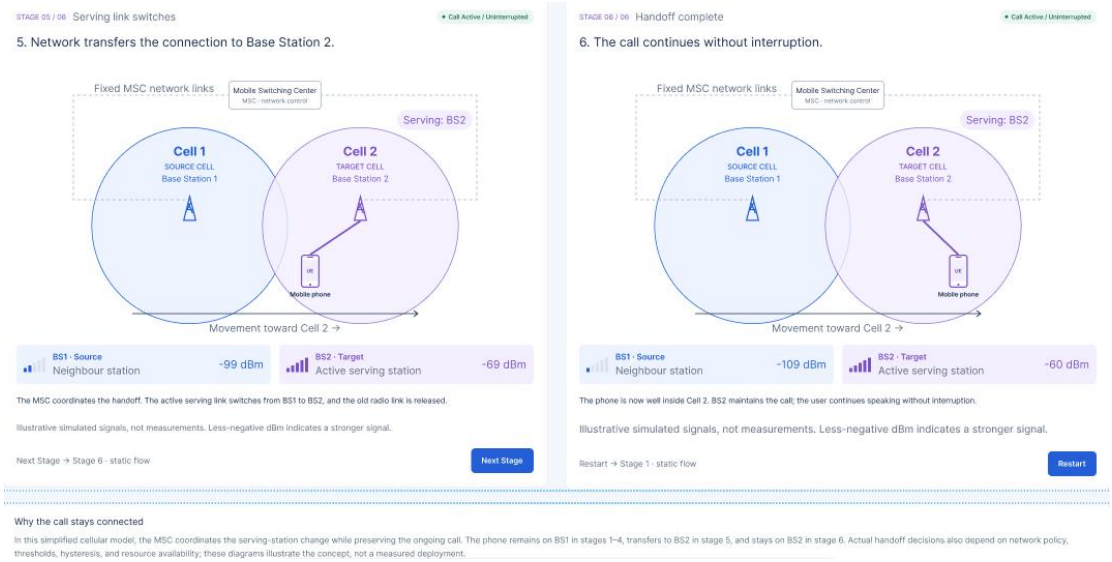
1. Serving base station: initial connection.
2. Target base station: destination connection.
3. Signal indicators: changing signal conditions.
4. Handoff decision: transfer point.
5. Active call: illustrates service continuity.

FIGMA LINK :

https://www.figma.com/design/3RcSkiK44stiFuvp277VjP/EXPERIMENT-NO-6?node-id=0-1&p=f&t=jRxRbwJtRBWP7omy-0

OUTPUT :


EXPER 6
Handoff scenario in a mobile network
Follow one ongoing call from the source cell to the target cell. Six consistent technical views make movement, radio conditions, and the serving connection easy to compare.
6 SEQUENTIAL DIAGRAM FRAMES
STATIC LAB DESIGN · Buttons and arrows describe intended flow only. No interactions, links, or prototype wiring are attached.
01 Sequence overview
Read left to right, then continue through the numbered frames below
1 Connected to BS1
2 Move to boundary
3 BS1 weakens
4 BS2 strengthens
5 Transfer to BS2
6 Call continues
Blue: source cell
Purple: target cell
Solid: active serving link
Fixed MSC network links
Phone movement
MANUAL FLOW REFERENCE / Stage 1: Start Handoff → 2. Stages 2–5: Next Stage → next numbered frame. Stage 6: Restart → 1. All controls are static; no prototype wiring is attached.
02 Handoff stages
Rows read left → right / 1–2, 3–4, 5–6
STAGE 01 / 06 Initial connection
Call Active / Uninterrupted
1. Mobile phone connected to Base Station 1.
Fixed MSC network links
Mobile Switching Center
MSC · network control
Serving: BS1
Cell 1
SOURCE CELL
Base Station 1
Cell 2
TARGET CELL
Base Station 2
UE
Mobile phone
Movement toward Cell 2 →
BS1 · Source
Active serving station
-62 dBm
BS2 · Target
Neighbour station
-108 dBm
The phone is well inside Cell 1. BS1 serves the active call; BS2 is a weaker neighbouring station.
Illustrative simulated signals, not measurements. Less-negative dBm indicates a stronger signal.
Start Handoff → Stage 2 · static flow
Start Handoff
STAGE 02 / 06 Movement begins
Call Active / Uninterrupted
2. Phone moves toward the boundary between cells.
Fixed MSC network links
Mobile Switching Center
MSC · network control
Serving: BS1
Cell 1
SOURCE CELL
Base Station 1
Cell 2
TARGET CELL
Base Station 2
UE
Mobile phone
Movement toward Cell 2 →
BS1 · Source
Active serving station
-74 dBm
BS2 · Target
Neighbour station
-98 dBm
As the phone moves right, BS1 signal decreases and BS2 signal improves. The serving connection remains with BS1.
Illustrative simulated signals, not measurements. Less-negative dBm indicates a stronger signal.
Next Stage → Stage 3 · static flow
Next Stage
STAGE 03 / 06 Source signal falling
Call Active / Uninterrupted
3. Signal from Base Station 1 weakens.
Fixed MSC network links
Mobile Switching Center
MSC · network control
Serving: BS1
Cell 1
SOURCE CELL
Base Station 1
Cell 2
TARGET CELL
Base Station 2
UE
Mobile phone
Movement toward Cell 2 →
BS1 · Source
Active serving station
-86 dBm
BS2 · Target
Neighbour station
-91 dBm
Near the cell edge, the source signal has weakened. The network evaluates neighbouring coverage while BS1 still carries the call.
Illustrative simulated signals, not measurements. Less-negative dBm indicates a stronger signal.
Next Stage → Stage 4 · static flow
Next Stage
STAGE 04 / 06 Target signal preferred
Call Active / Uninterrupted
4. Base Station 2 signal becomes stronger.
Fixed MSC network links
Mobile Switching Center
MSC · network control
Serving: BS1
Cell 1
SOURCE CELL
Base Station 1
Cell 2
TARGET CELL
Base Station 2
UE
Mobile phone
Movement toward Cell 2 →
BS1 · Source
Active serving station
-93 dBm
BS2 · Target
Neighbour station
-78 dBm
In the overlap region, BS2 now offers the stronger signal. The network prepares the target connection; BS1 remains the serving station.
Illustrative simulated signals, not measurements. Less-negative dBm indicates a stronger signal.
Next Stage → Stage 5 · static flow
Next Stage



Result:

Hence the simulated a handoff scenario in a mobile network and visualized the process using Figma.